



Towards an explainable machine learning model to reduce readmission risks for diabetes patients[☆]

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ABSTRACT

Objective: Hospital readmission of Diabetes patients is a persistent burden on the healthcare industry. Artificial Intelligence (AI) based Machine Learning (ML) techniques offer the potential to predict readmission rates and related risk features for diabetic patients. However, complex machine learning-based solutions are often not explainable and hard to understand for the relevant parties. To this end, this study designs and implements an explainable model to predict readmission rates and identify the risk factors associated with readmission in patients with diabetes.

Methods: The model employs various explainable visualization techniques, including the permutation importance plot, partial dependence plot (PDP), SHapley Additive exPlanations (SHAP), and interpretable classifiers on a publicly available dataset from US hospitals.

Results: The bagging random forest model yields the best results, achieving 89% accuracy and 67% precision.

Conclusion: The explainability visualization techniques reveal that the number of inpatient admissions and emergency visits in a year is the two most critical risk factors for the readmission rate of diabetic patients.

1. Introduction

Diabetes remains a growing global health concern with significant implications for healthcare systems. According to the International Diabetes Federation (IDF), approximately 463 million adults were living with diabetes worldwide in 2019, a number projected to increase to 700 million by 2045 [1]. In the United States alone, the Centers for Disease Control and Prevention (CDC) reported that as of 2021, 38.4 million people — about 11.6% of the population — have diabetes, including 8.7 million undiagnosed cases [2,3].

Beyond prevalence, diabetes is associated with high hospital utilization and readmission rates. Studies show that diabetic patients are readmitted within 30 days at rates ranging from 14% to 20%, depending on comorbidities and complications [4,5]. A landmark analysis of over 100,000 inpatient records revealed that HbA1c levels and insulin therapy status significantly influenced the likelihood of readmission [5]. These statistics underscore the pressing need for predictive

tools that not only assess risk but also provide transparent explanations to inform clinical decision-making.

Recently, many researchers have shown a great interest in resolving complications related to diabetes patient readmissions by applying AI technologies. Therefore, the dataset “Diabetes 130 US hospitals for years 1999–2008: Diabetes-readmission” has been widely studied [4,6,7]. However, most studies have not addressed the class imbalance problem in this dataset. Moreover, the existing predictive models also lack interpretability and explainability [8]. As a result, health professionals are reluctant to use these models for patient diagnoses due to ethical and trust issues.

Several studies have analyzed the dataset to develop classifiers that predict features contributing to readmission risk. However, there is still a need for significant improvements in methodology and model optimization, as many lack interpretability and explainability. This study employs a novel approach to optimize the model by evaluating

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the impact of various features on the readmission rate of diabetic patients, thereby addressing ethical concerns and ensuring user trust.

To this end, this study addresses persistent issues to improve the existing models. The contribution of our study is as follows:

- We systematically address the class imbalance problem by evaluating the effectiveness of multiple resampling strategies — no sampling, random oversampling, and SMOTE — in the context of diabetes readmission prediction. While SMOTE is a well-established method, our contribution lies in rigorously comparing these techniques and validating SMOTE's performance advantage through empirical evidence tailored to this high-stakes clinical domain.
- We incorporate interpretable machine learning (ML) models, such as decision trees, and apply explainability techniques, including SHAP and PDP, to provide deeper insights into the outputs of black-box models. It will help health professionals to make informed decisions. In turn, they can efficiently develop personalized healthcare plans for diabetic patients, thereby improving their quality of life.

Our methodology involves integrating interpretable and black-box machine learning models to predict hospital readmissions among patients with diabetes. We combine performance benchmarking with post-hoc explainability techniques (SHAP, PDP, permutation plots) to ensure both accuracy and interpretability. The key stages include data preprocessing, model training, performance evaluation, and explainability analysis.

The rest of this paper is organized as follows. Section 2 reviews recent literature on diabetes-related readmission prediction and explainable machine learning. Section 3 details the methodologies used in the study. Section 4 presents the dataset, preprocessing, sampling strategies, and machine learning models used in our study. Section 5 presents the exploratory data analysis. Section 6 presents the experimental setup for the various machine learning methods. Section 7 presents the evaluation results, including performance metrics and interpretability analyses. Section 8 discusses key findings, implications, and limitations. Finally, Section 9 concludes the paper and outlines directions for future work.

2. Related work

Diabetes poses a growing global health burden. The International Diabetes Federation estimates that in 2019, approximately 463 million adults had diabetes worldwide—projected to rise to 700 million by 2045 [1]. In the US, the CDC's 2022 report indicates 38.4 million Americans (11.6%) living with diabetes—of these, 8.7 million remain undiagnosed [2,3]. It underscores the critical need for advanced predictive tools to preempt complications and rehospitalizations.

2.1. Systematic reviews and surveys

Several recent systematic or scoping reviews have synthesized the body of work on machine learning (ML) for predicting hospital readmissions. Askar et al. [9] reviewed 116 ML-based studies on hospital admissions and readmissions (70 specifically on readmissions). They found wide variability in datasets, features, preprocessing, algorithms, and evaluation metrics, and highlighted significant gaps in reporting quality, interpretability, code sharing, external validation, and handling class imbalance.

Ho et al. [10] conducted a scoping review of 43 U.S.-based studies on ML for hospital readmission prediction. They reported a median AUC of 0.68 (range, 0.50–0.90), indicating a strong reliance on tree-based neural networks, regularized logistic regression, and SVM models. However, they noted inconsistent use of validation methods and limited transparency in model explanations.

A recent review evaluated machine learning (ML) applications across general internal medicine readmissions, finding AUCs typically ranging from 0.65 to 0.85. Still, few studies incorporated explainability tools or explicitly studied the impact of sampling methods on performance [11].

These reviews collectively identify gaps in comparative analysis of oversampling techniques (e.g., SMOTE, ROS), limited deployment of interpretable models in readmission contexts, and insufficient validation on real-world or imbalanced datasets. Our work aims to address these issues by integrating multiple sampling techniques and explainability methods into a single framework.

Our study addresses these gaps by systematically comparing sampling techniques, employing both performance metrics and interpretability analyses, and ensuring methodological transparency through the availability of code and data (see Table 1).

2.2. Explainable AI in diabetes and healthcare

The past few years have seen significant advancements in applying XAI techniques to diabetes ML models:

- **Ahmed et al. (2024)** combined SHAP and LIME on extensive diabetes cohorts, reporting 86% accuracy with interpretable feature influences [12].
- **Wang et al. (2023)** delivered an ensemble pipeline with SHAP-driven transparency for diabetes diagnosis [13].
- **Allani (2025)** developed an interactive Dash app integrating SMOTE sampling with SHAP/LIME to aid clinicians in diabetes risk prediction [14].
- **Claire-Del Granado et al. (2024)** used SHAP and LIME to identify key risk profiles in post-transplant readmission cases, emphasizing interpretability [15].

2.3. Interpretable machine learning approaches

Foundational and domain-specific work has advanced transparent ML models:

- **Bastani et al. (2017)** introduced model extraction to approximate black-box models with decision trees in medical inference tasks [16].
- **Luo (2018)** proposed a full interpretation pipeline for Type 2 diabetes predictions from EMR data [17].
- **ElShawi et al. (2020)** benchmarked local explanation methods (SHAP, LIME, Anchors, MAPLE) in healthcare contexts [18].
- **Kutlu et al. (2024)** employed SHAP to interpret XGBoost predictors in diabetes risk assessment [19].
- **Murdoch et al. (2019)** provided a comprehensive survey distinguishing interpretability and explainability in ML [20].

2.4. ML for hospital readmission prediction

A variety of studies have addressed readmission risk prediction, focusing on model performance and feature insights:

- **Pham et al. (2019)** employed ensemble models and clustering to predict diabetic readmission using a US hospital dataset [6].
- **Alloghani et al. (2019)** implemented various machine learning models, including decision trees and logistic regression, to profile readmissions [21].
- **Alamer et al. (2019)** validated two 30-day readmission prediction algorithms in diabetes patients, underscoring model comparison [4].

Table 1
Summary of key ML and XAI studies in diabetes and readmission (2017–2025).

Year	Study	Method	Contribution
2017	Bastani et al. [16]	Model extraction	DT surrogate for black-box
2018	Luo [17]	EMR interpretation pipeline	Full explainability system
2020	ElShawi et al. [18]	SHAP/LIME comparison	Local interpretability benchmark
2023	Wang et al. [13]	Ensemble + SHAP	Pipeline with transparency
2025	Ahmed et al. [12]	RF + SHAP/LIME	Large-scale diabetes interpretability
2024	Alnazari [15]	SHAP/LIME	Hospital readmission insights
2025	Allani [14]	SMOTE + SHAP app	Interactive XAI for risk
2019	Pham et al. [6]	Ensemble readmission	Clustered risk modeling
2019	Alloghani et al. [21]	Multiple models	Readmission profiling
2019	Alamer et al. [4]	Algorithm validation	Compared prediction methods

2.5. Gaps and our contributions

Although numerous studies apply ML and XAI to diabetes and readmission risk, important gaps remain:

1. Lack of comprehensive comparison among class imbalance strategies (none, oversampling, SMOTE) in healthcare-specific predictive models.
2. Limited exploration of how sampling decisions affect interpretability across model types.
3. Sparse evaluation of interactive XAI workflows grounded in clinical readmission datasets.

Our work addresses these gaps by:

- Systematically benchmarking imbalance corrections alongside explainability.
- Contrasting interpretable vs. black-box models under different sampling schemes.
- Demonstrating both predictive performance and interpretability in hospital readmission scenarios.

3. Methodologies

3.1. Explainable vs. Interpretable machine learning

In the context of healthcare, the transparency of machine learning (ML) models is crucial for fostering trust among clinicians, patients, and stakeholders. While the terms *explainable machine learning* and *interpretable machine learning* are often used interchangeably, they refer to distinct yet complementary concepts.

Interpretable Machine Learning (IML) refers to models whose internal structure and decision-making logic can be readily understood by humans. For instance, models such as decision trees or linear regression are considered inherently interpretable because their predictions can be directly traced back to input features in a comprehensible manner.

Explainable Machine Learning (XML), on the other hand, involves the use of techniques to provide post-hoc explanations for the decisions made by complex or black-box models, such as random forests or neural networks. These models may achieve higher accuracy but lack transparency. Explainability techniques, such as SHAP (SHapley Additive exPlanations), permutation importance, and partial dependence plots (PDPs), are used to interpret the behavior of these models.

Although both IML and XML aim to improve model transparency, they serve different roles. Interpretable models offer transparency by design, while explainable models require additional tools to make their decisions understandable. In this work, we combine both approaches: interpretable models for baseline transparency and explainable methods to elucidate predictions from high-performing black-box models.

In this study, we refer to models like decision trees as interpretable models and to techniques like SHAP and PDP that provide insight into complex models as explainable methods.

3.2. Study design

In this research, we focus on interpretable machine learning (ML) algorithms, such as decision trees (DT) and random forests (RF). We also experiment with other ML models, such as KNN, XGBoost, QDA, Logistic Regression, and AdaBoost, to compare their performance.

Since this research focuses on the explainability and interpretability of the models, decision trees (DT) and random forests (RF) have been chosen, which are known for their transparency and interpretability. In addition, the other non-interpretable models, such as KNN, XGBoost, QDA, Logistic Regression, and Adaboost, were experimented with to compare their performance with the interpretable models.

Decision trees are highly suitable for classification and regression tasks due to their simplicity and ease of interpretation. On the other hand, Random Forests is an ensemble of decision trees, well-known for their ability to handle imbalanced data and for their interpretability.

As these classifiers are standard among the research community, we ask interested readers to explore [22]. The primary models are built using traditional machine learning (ML) practices established in the literature, as described in [23]. We combine various feature vector selection methods to improve the model's performance. Based on the parameters of the best-performing model, visualization methods (Permutation Plot, Partial Dependence plot, SHapley Additive exPlanations) are employed to enhance the model's interpretability and identify the essential features that determine the readmission rate outcome. To build trust and provide insight into model decisions, we supplement our predictions with SHAP and PDP visualizations, which explain how input features influence model outputs.

The workflow of our research methods is as follows:

- Data preprocessing and EDA to improve feature engineering for this dataset and visualize the data distributions to enhance data interpretability.
- Adopt variance threshold from the filter, chi_square, mutual information, ANOVA, F-value, and recursive elimination algorithm from wrapper to tune the number of features and improve the model's accuracy.
- Implement a range of machine learning models.
- Optimize each machine learning model involving a combination of hyperparameter tuning and feature selection.
- Select the best final model and visually analyze the model output to enhance interpretability.
- Use this interpretable machine learning prediction model to predict the risk of readmission rates for diabetic patients and identify risk features.

We selected a mix of interpretable and black-box machine learning algorithms to explore the trade-offs between model transparency and predictive performance. Decision Trees and Logistic Regression were chosen for their inherent interpretability, which is critical for clinician trust in high-stakes settings. Random Forest, AdaBoost, and XGBoost were included due to their ensemble learning capabilities, often yielding superior accuracy and robustness in structured datasets. K-nearest neighbors and Quadratic Discriminant Analysis (QDA) were evaluated for baseline comparison.

3.3. Feature selection

This paper adopts two standard feature selection methods: Filter (ANOVA, mutual information, variance threshold) and Wrapper for data processing [24]. The statistical filter method measures the correlation between characteristics and dependent variables, selecting the relevant ones. However, the high number of features in the dataset can be a disadvantage of this method, as the calculation time is significant and the model tends to overfit. On the other hand, the wrapper employs the cross-validation method to select variables based on their usefulness and importance during model training. Therefore, models that apply the Wrapper method often exhibit better performance.

3.4. Visualization

We use various explainability techniques to interpret the behavior of our black-box models. These are Permutation Plots, Partial Dependence Plot (PDP), 2D PDP, and SHapley Additive exPlanations (SHAP). The details of each of these methods are out of the scope of this paper. Instead, we refer readers to [25] for more information on such practices.

4. Data collection and analysis

4.1. Raw data

The dataset was obtained from Humberto Brando's Kaggle [26]. The Health Facts database is a nationwide data warehouse that collects extensive clinical records from hospitals across the United States [5]. From 1999 to 2008, the raw data collection covered clinical care data from 130 hospitals and integrated distribution networks in the United States. The retrieved data contains a total of 55 features. The patient must be hospitalized for diabetes, with a stay of 1 to 14 days, and undergo various laboratory tests and drug use for these data to be collected [26]. The initial data we utilize is Beata Strack's, which has 50 features after being extracted and processed [5]. Features with a high percentage of missing values, such as 'weight' and 'payer code,' were removed to maintain data integrity. One-hot encoding was applied to categorical features to prepare the dataset for model training. The dataset was split into training and testing sets using an 80:20 stratified sampling to preserve class distributions. All models were implemented using Python 3.9, with libraries such as scikit-learn (version 1.2), XGBoost (version 1.7), and SHAP (version 0.41). The training was conducted on a local machine equipped with an Intel Core i7 processor and 16 GB of RAM. Default hyperparameters were used unless otherwise stated, and 5-fold cross-validation was employed to ensure robustness.

4.2. Data preprocessing

We used a data frame to store the dataset. The dataset comprises 101766 rows, indicating samples, and 50 rows that contain 49 features and one label in the original dataset. We did not use any data imputation method, as the proportion of missing values in the dataset was very low (< 5%), except for "payer_code", "medical_specialty", and "weight", which had more than 50% missing values. Therefore, we removed those three variables, as the imputation methods for these variables may not be reliable and may introduce significant bias into any subsequent analyses or modeling due to the high proportion of missing values.

The features "encounter_id" and "patient_nbr" only indicate the ID of the patient and the encounter. Therefore, they will not affect the model prediction. As a result, we dropped these two features. We also observed that many nominal features contained too many integer identifier categories. Therefore, we reduce the number of such categories to avoid high cardinality, overfitting, or sparse data problems.

Then, we applied the categorical binning technique to reduce the number of unique categories in the categorical variables presented in the data. For the feature "discharg_disposition_id", which initially contained 26 distinct categories, we combined them into 9 categories. For the feature "admission_type_id", which initially had 8 different categories, we combined them into 2 categories. For the feature "admission_source_id", originally included 17 distinct categories, we combined them into 6 categories. The features "diag_1", "diag_2", and "diag_3" originally contained 848, 923, and 954 distinct categories, respectively. We combined all of them into 9 categories. Then, we found that some nominal features include text categories. Thus, we converted the text categories to integer identifiers. Later, we dropped all the rows that contained missing values. Finally, we observed that "citoglipton", "examide", "glimepiride-pioglitazone", and "metformin-rosiglitazone" contained the same value for all rows, which means they do not affect model prediction. Thus, we dropped these 4 features. After the data preprocessing step, the dataset contained 99492 rows and 41 columns, which comprised 40 features and 1 label. Then we set the 40 features as our "x" and the 1 label as our "y".

We used MinMaxScaler to normalize numerical features between 0 and 1, thereby preserving feature relationships and ensuring compatibility with distance-based algorithms, such as k-nearest Neighbors (KNN). Features with excessive missing values or low predictive relevance were removed based on domain knowledge and exploratory analysis. One-hot encoding was chosen to handle categorical variables without introducing ordinal bias. Finally, we refer interested readers to our GitHub code for more information about our combination methods [27].

4.3. Class imbalance

The dataset had a significant data imbalance issue. For example, 88.77% of the samples belonged to label 0, and only 11.23% of the samples belonged to label 1. This problem leads to the models having good predictive performance on label 0 since this label contains many more samples than the other label. On the other hand, the models exhibit poor predictive performance on label 1, as this label comprises only a small number of samples.

Therefore, we adapted the Synthetic Minority Oversampling Technique (SMOTE) specified in the sklearn-library [28] to overcome this problem. Although SMOTE is not a novel technique, its effectiveness can vary significantly depending on the dataset and domain. In this study, we demonstrate that SMOTE offers substantial improvements in both recall and F1-score compared to other methods, such as random oversampling and no sampling. These improvements are particularly significant in healthcare applications, where false negatives (i.e., missed high-risk readmissions) can have severe consequences. Therefore, our contribution is not the method itself but rather a comprehensive, domain-specific validation of SMOTE's suitability for healthcare machine learning applications.

We report accuracy, precision, recall, and F1-score to assess model performance. In the healthcare context, precision and recall are critical due to the high cost of false negatives (i.e., missed readmission risks). Therefore, we prioritize models with balanced precision-recall trade-offs.

4.4. K-fold validation

This study employs the K-fold cross-validation method to enhance overall accuracy and mitigate the model's overfitting issue. K-fold cross-validation divides the training set data into K mutually exclusive subsets of equal size. The confusion matrix was applied to evaluate models. The details are in the Results section. We adopted the Pearson correlation coefficient to explore the relationship between different features using Python's seaborn library [29]. We observed a strong correlation between the time spent in the hospital and the number of medications prescribed, which is pronounced. Additionally, the analysis reveals a strong correlation between the number of medications and the number of procedures.

4.5. Evaluation

We report accuracy, precision, recall, and F1-score to provide a balanced view of model performance. In the healthcare domain, recall is especially critical, as it reflects the model's ability to correctly identify patients at risk of readmission—minimizing false negatives that could lead to delayed intervention. Precision helps ensure that alerts raised by the model are meaningful and actionable, reducing alarm fatigue. F1-score balances the trade-off between these two, while accuracy gives an overall correctness measure.

5. Exploratory data analysis

We apply EDA to find the link between gender, race, age, and the number of distinct re-recordings. We also analyze the relationship between the number of medications used during the hospital stay and the average number of drugs. The ethnic distribution is predominantly white and African American because these data are from the United States. Readmission rates did not differ significantly between men and women. Most of the patients are between the ages of 50 and 90.

We also observed the relationship between the number of medications used during the hospital stay and the average number of medications. It is observed that as the length of hospital stay increases to the longest 14 days, the average dosage of the medication gradually increases. The peak of the total dosage occurred on the third day, and most patients showed improvement by then. Afterward, they were discharged following a three-day hospitalization.

We also analyzed the relationship between the average length of hospital stay and the number of different medications, as well as the total number of procedures and laboratory tests. The number of surgeries and laboratories peaked on the third day of hospitalization.

6. Experimental setup

6.1. Feature selection

We have used the feature selection method in our models to facilitate the selection of the most relevant features, improving the model's accuracy and interpretability. We apply four different filter feature selection methods: the "VarianceThreshold" function, the "SelectKBest" function with the chi-square statistic, the "SelectKBest" function with the ANOVA F-value, and the "SelectKBest" function with the Mutual Information metric. We use different numbers of features for each model to find the optimal choice based on performance metrics such as accuracy, precision, recall, or F1 score. The details are explained in the subsections below.

6.2. KNN

To implement the KNN model, we initially set the parameter "n_neighbors" to 5, the default value. Next, we used the training data set to fit the model. Then, we used the testing set to see the model's predictive performance. Then, we tried different values of "n_neighbors" (1, 3, 7, 10, 15, etc.) to fit the model several times and checked the predictive performance. However, the value of "n_neighbors" did not significantly affect the prediction. Thus, we kept the "n_neighbors" as the default value. The other parameters were also not crucial to the prediction. Then, we focused on doing the feature selection section. We implemented four different filter feature selection methods: the "VarianceThreshold" function, the "SelectKBest" function with the chi-square statistic, the "SelectKBest" function with the ANOVA F-value, and the "SelectKBest" function with the Mutual Information metric.

6.3. Logistic regression

We applied the chi-square algorithm from the filter method with $k = 23$ to extract the 23 features with the highest weights for training the logistic regression. On the other hand, recursive feature elimination was used to select the 3 features ($n=3$) with the highest weights to train the model. Both feature selections apply the SMOTE algorithm to the training data, oversampling the class with fewer samples.

6.4. QDA

After going through the data preprocessing, which is roughly the same as other methods, the QDA data utilizes the previously processed data. However, compared with LDA, QDA has fewer restrictions. For example, there are no solver restrictions and no shrinkage restrictions. Therefore, QDA requires fewer parameter adjustments. The feature selection method uses chi-square, variance, and mutual information. In variance, set the value of k from 0.01 to 0.2. When it exceeds 0.2, the accuracy will be significantly reduced. In the remaining two methods, we set the value of k from 1 to 20, with an interval of 1. After 20, the accuracy rate began to drop significantly. The specific results will be described in detail in the results section below.

6.5. XGBoost

We trained the same parameters on data specified by different features to identify a set of datasets with the highest accuracy and precision, and derived the appropriate features. We selected three feature selection algorithms: chi-square, variance threshold, and ANOVA f-value. For the variance method, we tested 50 groups, varying k from k to 0.50 in increments of 0.01, and recorded the accuracy and precision values. Because when k exceeds 0.50, the precision value of the data decreases, we choose a maximum k value of 0.50. We employed the same training strategy for the remaining two feature selection methods, testing 43 sets in the k -value range from 1 to 43, with the same recording accuracy and precision values. Finally, we combined the comparisons and selected the highest set of precision and recall with the same feature names in the dataset.

6.6. AdaBoost

The study adjusted the parameters in the trade-off between $n_estimator$ and $learning_rate$ in the AdaBoost algorithm. The $n_estimator$ was increased to 200 (default 50), and $learning_rate$ was set to 0.5 (default 1.0), and a Gini coefficient-based decision tree was used as the primary classifier. Additionally, the maximum depth in the decision tree was set to 1 to reduce computational cost. On the other hand, the PCA results showed that the top 29 features, which explained the highest variance, accounted for 80% of the variance. Therefore, we set five feature subsets, including 40 features without feature selection and four feature subsets with $k = 18, 23, 29$, and 33 using chi-square feature selection. We experimented with five different models to select the best-performing subset of features.

6.7. Decision tree

The unified data are preprocessed first, and then the `train_test_split` function is used to divide the dataset into two parts: a training dataset and a test dataset. However, we identified a class imbalance issue throughout the entire dataset process. Therefore, we choose an over- or undersampling method, such as SMOTE/OneSideSelection, to address this problem and balance the data amount in the training dataset. For the model selection method, we choose the variable mainly for the filter method, including variance, chi-square, analysis of variance F-value, and mutual information. We then employed this method to train the decision tree model, but did not achieve improved performance, particularly in terms of the recall score.

Table 2

The predictive results of the random forest model integrated optimal approaches for handling the class imbalance problem.

Method	Function	Accuracy	Precision
Over-sampling	SMOTE after parameter tuning	0.89	0.44
Under-sampling	OneSidedSelection after parameter tuning	0.89	0.56
Combination of over-and under-sampling	SMOTETomek after parameter tuning	0.89	0.42
Ensemble of samples	BaggingClassifier with random forest classifier after parameter tuning	0.89	0.67

6.8. Random forest

We applied the `train_test_split` function to split our “x” and “y” into the training and testing sets. Note that the parameter “stratify” in `train_test_split` was set to “y”. It is because we found that the class labels in “y” were extremely imbalanced, and putting “stratify” equal to “y” can make the training set and testing set have the same proportion of class labels as the proportion of class labels in the original dataset to reduce the class imbalance problem. After training with the training set, the base random forest model, with all parameters set to their default values, achieves an accuracy of 0.89 and a precision of 0.53 for prediction.

Then, we employed the 10-fold cross-validation for the grid search function to perform parameter tuning for the random forest model. In the “param_grid” dictionary of string to sequence, we set “[‘gini’, ‘entropy’]” for the parameter “criterion”, “[50, 100, 150, 200]” for the parameter “n_estimators”, “[‘auto’, ‘log’, 40]” for the parameter “max_features”, “[10, 20, 30, 40, 50, 60, 70, 80, 90, ‘None’]” for the parameter “max_depth”, “[2, 5, 10, 15, 20]” for the parameter “min_samples_split” and “[1, 2, 5, 10, 15]” for the parameter “min_samples_leaf”.

7. Results

7.1. Random forest

We apply a 10-fold cross-validation grid search for the RF model. The model achieves an accuracy of 0.89 and a precision of 0.55 with the top 35 most important features. We experimented with four random forest models using different feature numbers. We found that the model that selected the 32 most important features was the best. Therefore, we trained the model with those features and achieved an accuracy of 0.89 and a precision of 0.61.

We experimented with four approaches to address the class imbalance problem: oversampling, undersampling, a combination of oversampling and undersampling, and an ensemble of samplers. In addition, we experimented with multiple functions in each approach. Then, we selected the best function corresponding to each approach: Smote, OneSidedSelection, Neighborhood-Cleaning Rule, and SMOTETomek (Table 2).

Table 2 reveals that the model trained with the after-SMOTE training data had worse predictive performance; the optimal random forest model trained with the after-SMOTE data decreased precision from 0.61 to 0.44.

For the model that employed the OneSidedSelection method, the precision improved from 0.51 to 0.56. The precision of the random forest model improved by only 0.01 from 0.41 using the SMOTE-Tomek method. On the other hand, we applied the bagging classifier model, utilizing random forest models as the weak classifiers, to overcome the class imbalance problem. We modified the default parameter “base estimator” in the bagging classifier model from the decision tree classifier

to the pre-tuning random forest classifier. As a result, the precision value increased to 0.65 from 0.61. When the after-tuning random forest classifier was selected as the base estimator, the precision increased to 0.67, the best result among all other models.

7.2. Decision tree

The results of many selected tree models with different features did not change significantly, indicating that variable selection did not particularly improve the model’s performance. When the optimal hyper-parameters of ‘criterion’: ‘entropy’, ‘max_depth’: 5, ‘min_samples_leaf’: 100, and ‘min_samples_split’: 50 were selected, the model achieved its highest precision from 0.15 to 0.37, and accuracy from 0.77 to 0.89 (Table 3).

7.3. Other classifiers

Table 3 shows the results from KNN, QDA, XGBoost, Logistic Regression, and AdaBoost. KNN models achieve an accuracy of 0.88 and a precision of 0.19 when all 40 features are used. The results did not improve after applying the chi-square test, ANOVA F-value, and Mutual Information. Thus, we kept the initial result as the best result for the KNN model. The improvement of the variance method to QDA is minimal, and the accuracy is only about 0.53. Chi-square and mutual information methods can improve the performance of QDA.

The performances of the different XGBoost models show immunity to the feature selection method. It maintains a steady accuracy of 0.89 regardless of feature selection. The Logistic Regression model achieved a poor result, with an accuracy of 0.66 and a precision of 0.16. The AdaBoost model achieves an accuracy of 0.89 and a precision of 0.35.

7.4. Visualizations

7.4.1. Permutation plot

The permutation importance ranked the features in the prediction model employing the random forest algorithm. We observed that the quantitative variable “number_emergency” is the most crucial feature, with a higher weight than the second-ranked feature “number_outpatient” and the third-ranked feature “number_inpatient”. On the other hand, the ordinal variables such as “diag_3” and “diag_1” do not seem to occupy a more significant weight (Fig. 1). Meanwhile, there was no statistically significant effect on the features of any diabetes medication.

7.4.2. PDP (partial dependence plot)

Following the random forest model’s ranked feature importance and permutation plot, the PDP was used to visualize the heavily weighted features and medications that may impact the readmission rate, where the blue shading indicates the confidence interval.

The number of inpatients shows the most substantial relationship with readmission rates, with the steepest slope among all the features

Table 3
Results of other classifiers.

Classifier	Feature selection	Accuracy	Precision
KNN	Variance Threshold	0.90	0.15
Logistic Regression	Recursive Feature Elimination	0.66	0.16
Decision Tree	Mutual Information	0.79	0.15
AdaBoost	Chi-square	0.89	0.35

Weight	Feature
0.0002 ± 0.0002	number_emergency
0.0002 ± 0.0001	number_outpatient
0.0002 ± 0.0001	number_inpatient
0.0002 ± 0.0002	number_diagnoses
0.0001 ± 0.0002	diag_3
0.0001 ± 0.0002	diag_1
0.0000 ± 0.0001	glipizide
0.0000 ± 0.0002	change
0.0000 ± 0.0002	race
0.0000 ± 0.0002	num_procedures
0.0000 ± 0.0002	gender
0.0000 ± 0.0001	admission_source_id
0.0000 ± 0.0001	glyburide
0.0000 ± 0.0001	diabetesMed
0 ± 0.0000	acarbose
0 ± 0.0000	examide
0 ± 0.0000	metformin-pioglitazone
0 ± 0.0000	nateglinide
0 ± 0.0000	glipizide-metformin
0 ± 0.0000	glyburide-metformin
... 24 more ...	

Fig. 1. Result of permutation plot.

(Fig. 2). Furthermore, we observed that when the total number of diagnoses ranges from 2 to 4, the readmission rate decreases as the number of diagnoses increases. Conversely, when the number exceeds 4, the readmission rate increases in proportion to the total number of diagnoses. However, the boundary value for the variable fluctuated around 5%. Therefore, it was not a significant risk factor for the admission rate to decrease as the number of diagnoses increased.

We observed that the readmission rates among older generations in the samples decreased. Although the trend increased at ages older than 60 years, the negative readmission rates still indicate that age appears to be unrelated to readmission rates.

The number of emergency room visits was strongly correlated with the readmission rate, and the growth was significant. The more emergency room patients visited within a year, the higher the probability of being readmitted (Fig. 3). The number of medications did not interfere significantly with readmission rates. When patients had fewer than three medications, increasing the number of medications did not result in a reduction in readmission rates. When the number of medications exceeded three, readmission rates increased.

7.4.3. 2D PDP

Correa, Rodriguez, and Nappe [30] stated that glipizide might interact with other diabetic medications, such as insulin, to induce hypoglycemia. Therefore, the importance of a single feature may not fully explain the model. The interaction between multiple features may potentially increase the risk of readmission in patients with diabetes.

We observed that using either repaglinide or insulin increases readmission rates, with the highest probability of readmission occurring when both insulin and repaglinide are used.

We observe the interaction of the high-weighted features of the number of inpatients per year and age. The probability of readmission

rates is highest among patients who have six admissions per year and are under 85 years old. The number of inpatients is found to be a heavily weighted feature. The readmission rate would depend solely on the number of inpatients rather than their age.

When the number of medications is 2, there is a significant decrease in readmission rates. However, a high number of medications and emergency room visits both lead to increased readmission rates. In contrast, the number of emergencies has a marginally higher impact than medications.

We also analyzed the interaction of repaglinide with other medications, which increased readmission rates but had a less significant impact on the number of inpatients (see Fig. 4).

7.4.4. SHAP (SHapley Additive exPlanations)

The SHAP summary plot shows that the number of inpatient visits and time spent in the hospital are the most influential features in predicting readmission. It aligns with clinical understanding, where repeated admissions or prolonged hospital stays often indicate deteriorating patient conditions. Partial Dependence Plots further support these findings, showing a positive correlation between readmission probability and inpatient visits.

Each row represents a feature in the SHAP plot, and the horizontal coordinates are the SHAP values (Fig. 5). Each dot represents a sample, with red indicating a high feature value and blue the opposite. It is noticeable that the number of inpatients remains the most significant feature and is negatively correlated with the probability of readmission, meaning that a lower number of admissions results in a lower likelihood of readmission.

Age is ranked higher in importance in the SHAP plot than in the PDP or permutation plot, with older age leading to a lower probability of readmission rates. On the other hand, if the patient is considered too young, they will be more likely to be readmitted.

It was found that the number of outpatients did not align with the number of inpatients, with a high number of outpatients having a higher probability of readmission. Still, an excessive number of outpatients had a lower inpatient readmission rate.

8. Discussion

In this paper, an interpretable predictive model is developed that identifies the most critical risk factors and predicts the risk of readmission for diabetic patients according to these factors.

Some studies find that the decision tree-based model performs better on diabetic readmission data. For example, Zhu et al. [31] employed a decision tree-based C4.5 algorithm to build a predictive model that overcame missing values and achieved an accuracy of 74.5%. In addition, Tamin and Iswari [32] suggest that a modified random forest using the Gini coefficient is superior to other models. They found that the patient had been an inpatient at the hospital for a year. The number of lab procedures, the number of emergency visits, and the number of 29 disposition IDs were the risk features.

Our work selects the bagging random forest model as the final predictive model. Fig. 6 illustrates the best results from all the models used in this study. The bagging random forest model outperforms other models. The accuracy of linear regression, naive Bayes, and linear discriminant analysis is relatively low, only about 0.71. Although the accuracy of secondary discriminant analysis reaches 0.81, it still falls short of the ideal. KNN, ADB, XGB, DT, RF, and RF-bagging

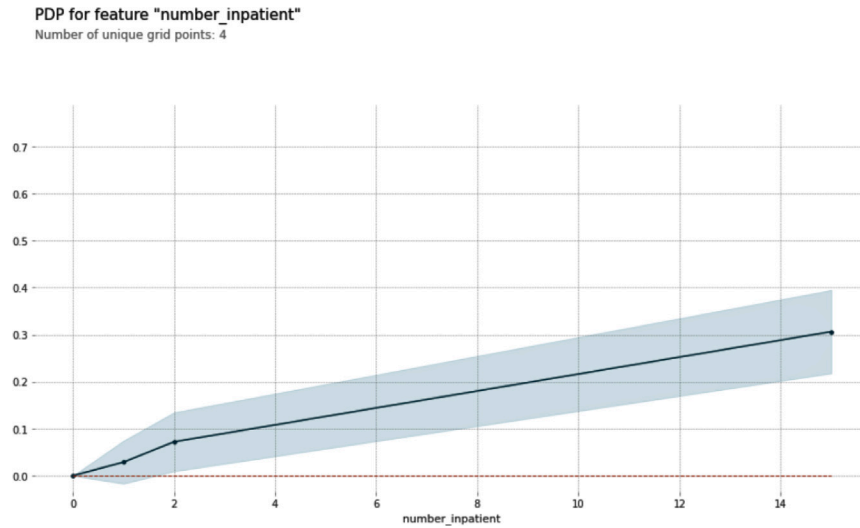


Fig. 2. PDP for the number of inpatients.

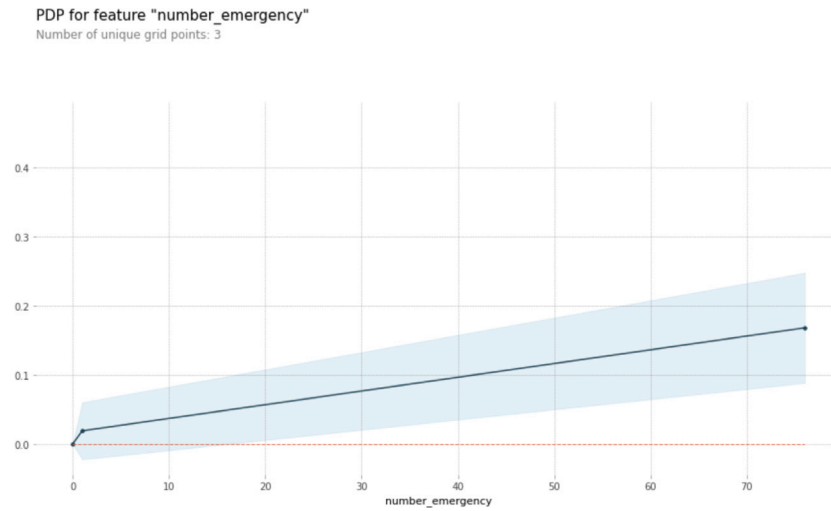


Fig. 3. PDP for number of emergency features.

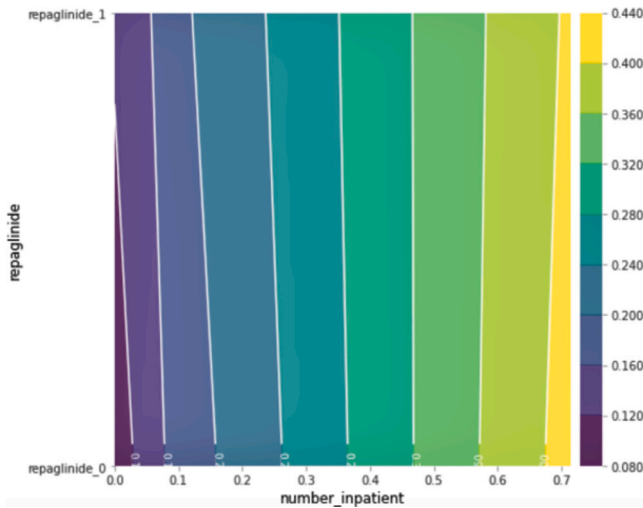


Fig. 4. 2D-PDP for number of inpatient & repaglinide.

accuracy are improved compared to previous models. The accuracy of RF-bagging is the highest. At the same time, RF bagging also has the best precision value. Its accuracy of 0.89 indicates that it outperforms other classification models based on the C4.5 algorithm [31], modified random forest [32], and other classification models in this paper. While its precision of 0.67 on the positive class also indicates that it outperforms other models.

The Bagging algorithm solves the class imbalance problem in the base model. It brings the model as close as possible to the lowest point of the generalization error, which complements previous research on the stability of decision tree models. It does not require feature selection and is straightforward to implement. To address its poor interpretability, we combine it with various visualization techniques, thereby improving its interpretability.

Gini coefficients-based feature ranking of RF shows that the number of procedures and medications were the two most important features [32]. However, the 2D PDP relationship plot in this study shows that these two features are less important than the number of inpatients per year and emergency visits. On the other hand, the permutation importance plot shows only seven features with weights of importance above 0.0001. The two most critical risk factors are the number of inpatients per year and the frequency of emergency visits.

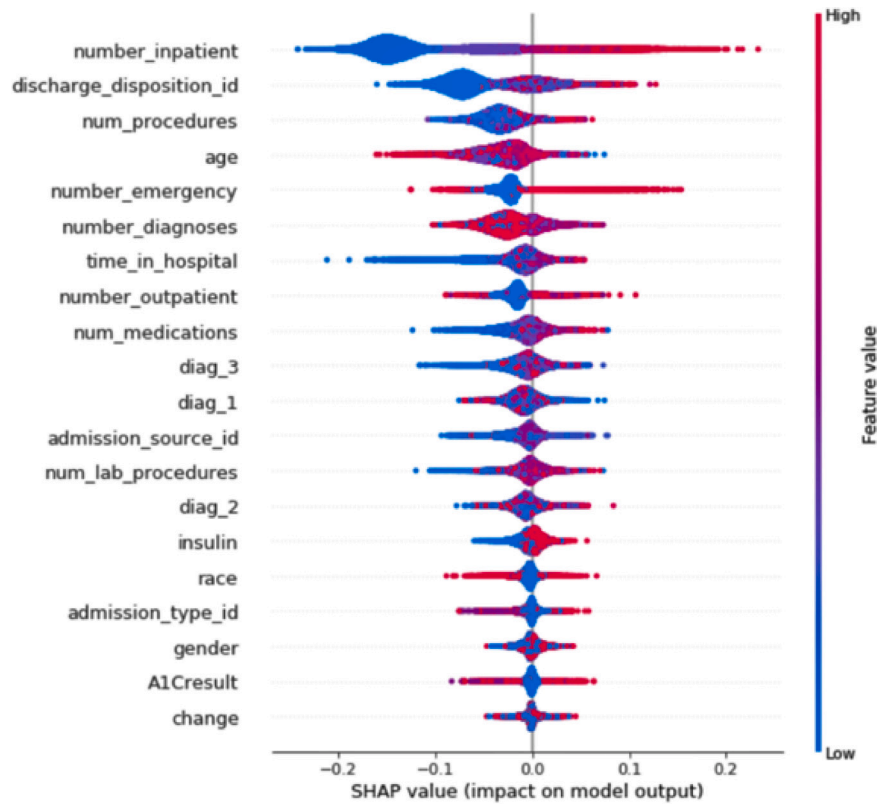


Fig. 5. SHAP Plot.

MODEL COMPARISON

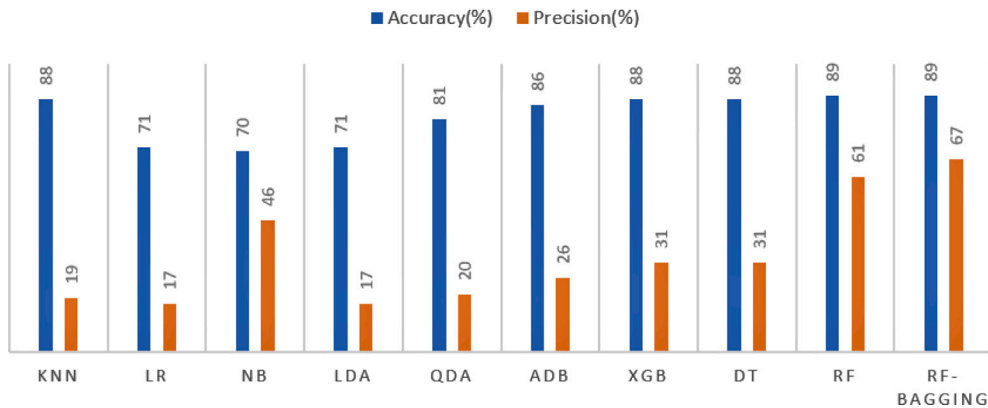


Fig. 6. Model comparison.

The results for PDP and 2D PDP are consistent with those of permutation importance, with the maximum values for both the number of inpatients in a year and emergency visits increasing readmission rates to or near 20% and showing stronger dominance compared to the other features. However, we found that the 6th-ranked feature, glipizide, had no significant impact. Thus, only the top 5 features of the permutation plot could provide better performance in predicting readmission rates.

The SHAP results indicate that the number of inpatients is the most significant risk factor, with discharge disposition ID, number of procedures, and age having a greater impact than emergency visits. However, the PDP results show that age is not strongly associated with the readmission rate. The SHAP model does not employ a bagging random forest model as the explainer. Therefore, a class imbalance issue exists. Only a few young people were predicted to be readmitted

(labeled as 1). The same problem exists for the number of procedures, and this paper does not consider disposition ID as an explainable risk feature, as it is an ordinal variable.

Combining the results of all the visualizations, we find that the number of inpatients in a year and emergency visits were the two most critical risk factors that explain the readmission rate of diabetic patients, as found by the authors of [32]. In addition, the PDP and 2D PDP in this paper generated interpretable outputs, thereby enhancing the model's interpretability and facilitating the explanation of interactions and relationships between features, as well as feature importance plots. Finally, although the study by Strack et al. [5] suggests that HbA1c is a significant predictor. However, this feature did not significantly impact output in this study's random forest and SHAP models.

Table 4

Comparison of our work with selected studies on diabetes-related readmission prediction.

Study	Dataset	Sampling	ML models	Explainability	Interpretable model	Code public
Bojja et al. [33]	UCI	None	SVM, DT	No	Yes	No
Alloghani et al. [21]	EHR	None	ANN, DT, NB	SHAP	Yes	No
Pham et al. [6]	Clustered EHR	None	Ensemble, Clustering	No	No	No
Alamer et al. [4]	Hospital data	No	Logistic Regression	No	Yes	No
Strack et al. [5]	UCI	None	Logistic Regression	No	Yes	No
This Work	UCI	SMOTE, ROS	DT, RF, XGBoost	SHAP, PDP	Yes	Yes

8.1. Comparative analysis of related works

To further contextualize our contribution, Table 4 compares key recent studies in diabetes readmission prediction. While most studies employ standard machine learning (ML) techniques, few systematically evaluate sampling strategies or model explainability. Furthermore, interpretable models are seldom included, and performance is often reported without assessments of fairness or interpretability.

As shown in Table 4, our work:

- Benchmarks multiple sampling techniques (none, ROS, SMOTE), which are rarely evaluated in prior work.
- Includes both interpretable (e.g., decision tree) and black-box (e.g., XGBoost) models to balance performance and interpretability.
- Applies SHAP and PDP to enhance model transparency and user trust.
- Provides public code and reproducibility details for academic transparency.

8.2. Final remarks

While some studies have employed PCA dimensionality reduction or wrapper methods for feature selection, this study utilizes more comprehensive feature selection methods, including variance threshold from the filter, chi-square, mutual information, ANOVA F-value, and recursive elimination algorithm from the wrapper. Furthermore, we have independently tuned the feature selection algorithms for different classifiers and used tailored feature selection parameters. This approach has overcome the lack of homogeneity in feature selection among classifiers in previous studies. On the other hand, we employed cross-validation to select the appropriate parameters for all classifiers, which significantly improved their performance and increased our confidence in the algorithm. While [31] achieved 74.5% using a CS4.5-based decision tree model, our decision tree boosted the accuracy to 89%. In addition, this study compared the most commonly used machine learning classifiers. Therefore, the random forest model selected was more representative and more accurate. However, the robustness of the models has not been thoroughly investigated.

On the other hand, this study's three different visualization approaches enhance the interpretability of machine learning black box models and aid in explaining clinical theories. Xie et al. [34] presented that the combined use of repaglinide and insulin significantly increases the risk of disease in patients. Although Tamin and Iswari [32] have also demonstrated that repaglinide and hospital admissions per year influence readmission rates, the random forest model, being a black box, does not adequately explain this result. Whereas the PDP in this study demonstrated this trend (Fig. 2), the SHAP plot clearly shows the positive and negative correlations between features (Fig. 5).

8.3. Limitations

This paper improves the accuracy of the prediction model and fills the gap in the interpretability of previous studies. However, we acknowledge a few limitations. Firstly, we worked on a single dataset

and removed some features with missing values in the data processing section. Hence, the data size was limited. Secondly, although we tried multiple combinations of feature selection, there is still room for optimization. For example, the embedded algorithm can be applied to iterate over all features and then perform feature selection for models, including logistic regression. Thirdly, we only created PDP plots for some features, and there may be potential patterns for features that were ignored.

Additionally, there is a low false alarm rate and high recall, precision, and accuracy. Our focus was on correctly identifying many true positive cases while minimizing false positives, which could lead to unnecessary testing or treatment. Although we did not develop a new resampling algorithm, our findings highlight the practical utility of SMOTE in this clinical setting. Future work could explore advanced imbalance correction methods or hybrid sampling strategies specific to patient subgroups.

9. Conclusion and future works

In this paper, we developed a predictive model to analyze factors that may lead to readmission in diabetic patients and to predict their likelihood of readmission. We test our model using a dataset from 130 US hospitals that includes patients with diabetes from 1999 to 2008. We used a range of Machine learning classifiers and different feature selection methods, and compared their results. We also apply the Bagging classifier to oversample the dataset, thereby overcoming the significant class imbalance problem in this dataset that has not been previously addressed in similar studies. The bagging Random Forest model outperforms other classifiers with 89% accuracy and 67% precision. We apply several visualization techniques to analyze the significant effects of specific features on the readmission of diabetic patients. The two most critical risk factors are the number of inpatients per year and the frequency of emergency visits. In the future, we will apply this predictive model to an integrated dataset, such as the Pima Indians Diabetes Database and the Diabetes dataset from 130 US hospitals for 1999–2018, providing us with more samples and dimensions to validate the model. On the other hand, training the model using more complex neural networks and deep learning models may result in higher accuracy and require a more comprehensive eXplainable AI system.

CRedit authorship contribution statement

Changfeng Guo: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haoran Zhou:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ivan Miguel Pires:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Paulo Jorge Coelho:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Runzhe Tong:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Farnaz Farid:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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